

Effects of Anthropomorphic Design Cues of Chatbots on Users' Perception and Visual Behaviors

Citation for published version (APA):

Chen, J., Guo, F., Ren, Z., Li, M., & Ham, J. (2024). Effects of Anthropomorphic Design Cues of Chatbots on Users' Perception and Visual Behaviors. *International Journal of Human-Computer Interaction*, 40(14), 3636-3654. <https://doi.org/10.1080/10447318.2023.2193514>

Document license:
TAVERNE

DOI:
[10.1080/10447318.2023.2193514](https://doi.org/10.1080/10447318.2023.2193514)

Document status and date:
Published: 01/01/2024

Document Version:
Publisher's PDF, also known as Version of Record (includes final page, issue and volume numbers)

Please check the document version of this publication:

- A submitted manuscript is the version of the article upon submission and before peer-review. There can be important differences between the submitted version and the official published version of record. People interested in the research are advised to contact the author for the final version of the publication, or visit the DOI to the publisher's website.
- The final author version and the galley proof are versions of the publication after peer review.
- The final published version features the final layout of the paper including the volume, issue and page numbers.

[Link to publication](#)

General rights

Copyright and moral rights for the publications made accessible in the public portal are retained by the authors and/or other copyright owners and it is a condition of accessing publications that users recognise and abide by the legal requirements associated with these rights.

- Users may download and print one copy of any publication from the public portal for the purpose of private study or research.
- You may not further distribute the material or use it for any profit-making activity or commercial gain
- You may freely distribute the URL identifying the publication in the public portal.

If the publication is distributed under the terms of Article 25fa of the Dutch Copyright Act, indicated by the "Taverne" license above, please follow below link for the End User Agreement:

www.tue.nl/taverne

Take down policy

If you believe that this document breaches copyright please contact us at:

openaccess@tue.nl

providing details and we will investigate your claim.



Effects of Anthropomorphic Design Cues of Chatbots on Users' Perception and Visual Behaviors

Jiahao Chen, Fu Guo, Zenggen Ren, Mingming Li & Jaap Ham

To cite this article: Jiahao Chen, Fu Guo, Zenggen Ren, Mingming Li & Jaap Ham (2024) Effects of Anthropomorphic Design Cues of Chatbots on Users' Perception and Visual Behaviors, International Journal of Human-Computer Interaction, 40:14, 3636-3654, DOI: [10.1080/10447318.2023.2193514](https://doi.org/10.1080/10447318.2023.2193514)

To link to this article: <https://doi.org/10.1080/10447318.2023.2193514>



Published online: 05 Apr 2023.



Submit your article to this journal [↗](#)



Article views: 1608



View related articles [↗](#)



View Crossmark data [↗](#)



Citing articles: 9 View citing articles [↗](#)



Effects of Anthropomorphic Design Cues of Chatbots on Users' Perception and Visual Behaviors

Jiahao Chen^a , Fu Guo^a , Zenggen Ren^a, Mingming Li^a, and Jaap Ham^b

^aDepartment of Industrial Engineering, School of Business Administration, Northeastern University, Shenyang, China; ^bResearch Group of Human Technology Interaction, Department of Industrial Engineering and Innovation Sciences, Eindhoven University of Technology, Eindhoven, The Netherlands

ABSTRACT

Measurement of users' perception and visual behaviors to anthropomorphic design cues of chatbots can improve our understanding of chatbots and potentially optimize chatbot design. However, as two typical and basic features, how chatbot appearances and conversational styles jointly affect users' perception and visual behaviors remains unclear. Therefore, this study conducted an eye-tracking experiment to explore users' perception and visual behaviors. Results indicate that anthropomorphic appearances and human-like conversational styles jointly increased users' perception of chatbots' social presence, trust in chatbots, and satisfaction with chatbots. In contrast, on users' visual behaviors, such a joint effect was not found, although chatbots with higher anthropomorphic appearances and human-like conversational styles triggered more fixation counts and longer dwell time. These findings suggest that anthropomorphic appearance and human-like conversational style can improve users' perception and attract more visual attention to chatbots. These findings provide theoretical contributions and practical implications for relevant researchers and designers.

KEYWORDS

Chatbot appearance; conversational style; user perception; visual behaviors; eye-tracking

1. Introduction

With the rapid development and implementation of artificial intelligence (AI) technology, interactive chatbots are widely used on many e-commerce websites. Since chatbots are able to provide 7/24-h service for dealing with customers' questions, they can contribute to increasing engagement and reducing the response time in the conversation (Ashfaq et al., 2020). Additionally, the application of chatbots leads to higher customer satisfaction, lower costs, positive product attitudes, and greater purchase intentions (Chung et al., 2020; Xie et al., 2022).

Chatbots can interact with users, for example with customers on websites through a chat window. During such interactions, the design cues of chatbots play crucial roles in users' perception and experience. Although the chatbot is a computer-based system that interacts with users in natural language, they are often designed with anthropomorphic styles to make users feel that the interaction with the chatbot is more real and natural (Følstad & Brandtzaeg, 2020). An artificial social agent's (such as a chatbot's) level of anthropomorphism is defined as "the extent to which an agent acts and behaves like a human" (Meyer et al., 2016). Anthropomorphic cues of chatbots include visual appearances and conversational styles (Araujo, 2018; Go & Sundar, 2019). Users interacting with chatbots with anthropomorphic avatars and conversational styles may feel more

like they are interacting with a real person rather than with a machine (Go & Sundar, 2019). Additionally, these anthropomorphic design cues are able to enhance the emotional appeal of the website and increase user satisfaction (Etemad-Sajadi, 2016; Wang et al., 2007). However, Nowak (2004) argued that low anthropomorphic images of chatbots are considered more likable and credible than high anthropomorphic images. Ciechanowski et al. (2019) manifested that chatbots with too many anthropomorphic design cues may lead to an uncanny valley effect (a feeling of eeriness or revulsion towards an artificial social agent (Song & Shin, 2022)), especially for embodied agents. Still, existing research has found that anthropomorphic cues can affect users' perception without making which level of anthropomorphism had what effect.

Users' perception of chatbots is mainly affected by the visual appearance and conversational style. That is, earlier research suggested that anthropomorphic chatbots can positively affect perceived social presence (Liebrecht et al., 2020), perceived trust (Seitz et al., 2022), and satisfaction (Araujo, 2018). However, earlier studies primarily investigated separately the effects of visual appearances or the effects of conversational styles of chatbots on users' perception. No research considers the joint effect of these two factors, even though both of them are essential elements of chatbots. Moreover, users' perception is related to intrinsic and

unconscious information processing, which cannot be accurately measured only through the method of self-reporting (Luan et al., 2016). Various factors, such as peer pressure, time pressure, motivation or even unawareness, may lead to the users' unwillingness or inability to express their intrinsic thoughts (Guo & Zhang, 2020). In contrast, psychophysiological measures (Spelt et al., 2019), such as eye-tracking and electroencephalography (EEG), can observe users' physiological responses and bridge users' intrinsic emotional responses and subjective perception. In line with this, the current study employed eye-tracking combined with a subjective evaluation to explore the joint effect of anthropomorphic appearances and conversational styles of chatbots on users' perception and visual behavior.

In the human-computer interaction domain, the eye-tracking method has been widely used to evaluate website design and interface usability (Cao, Ding, et al., 2021; Wang et al., 2021). Unfortunately, despite the importance in this field, an eye-tracking method has not been adopted to assess chatbot design. Hence, the current study investigated the effects of anthropomorphic design cues of chatbots on users' perception and visual behaviors by adopting the eye-tracking method.

1.1. Users' perception of anthropomorphic design cues of chatbots

According to the media equation theory, people react to artificial agents in the same way as to other people, and this response is inevitable and automatic (Reeves & Nass, 1996). Additionally, the Computers Are Social Actors Paradigm (CASA) (Nass et al., 1996) suggested that users are more likely to self-disclose when computers follow the rules of human conversation. Correspondingly, chatbots could also make users feel like they are interacting with a human. Previous research indicated that agents with higher human-likeness provide more social cues and then lead to more social responses (Nass & Moon, 2000). That is, the more computers show human-like characteristics, the more social responses will be elicited (Nass & Moon, 2000). This research is also supported by the social agency theory (Wiseman et al., 2012) that anthropomorphic cues could trigger users positive social responses. When users interact with an agent that exhibits anthropomorphic visual cues or language styles, these features excite users to expect their interaction as a human-human interaction and lead to humanness perceptions (Jackson & Williams, 2021). Therefore, anthropomorphic cues play a crucial part in designing human-like chatbots for natural interaction. Indeed, earlier research has suggested that different manipulations of anthropomorphic cues can affect users' perception of chatbots (Go & Sundar, 2019). However, although earlier research found that anthropomorphic chatbots have effects on users' perception, it's still unclear which level of anthropomorphism had what effect.

In the current research, we focus on the influence of anthropomorphic cues of chatbots on three elements of chatbot perception: perceived social presence of the chatbot

(Tsai et al., 2021), the user's trust in the chatbot (Mostafa & Kasamani, 2022), and the user's satisfaction with the chatbot (Klein & Martinez, 2022). More specifically, social presence can be described as a psychological state that the intelligent agents are experienced as social actors (e.g., a real person) (Lee, 2004). Social presence is increased when users are exposed to agents with human-like features (Oh et al., 2018). Correspondingly, Qiu and Benbasat (2009) argued that anthropomorphic appearances of chatbots can create a feeling of social presence. Furthermore, trust can be argued to be essential for human-chatbot interaction. A crucial reason for this is that users' trust in the chatbot will affect the user's behavioral intentions (Benbasat & Wang, 2005). Indeed, Liebrecht et al. (2021) showed that chatbots with anthropomorphic design cues make users experience the interaction as more natural and increase the user's trust in the chatbot. Finally, user satisfaction is typically considered as the user's attitudes (i.e., the user's evaluation of the chatbot on a dimension from negative to positive) about the chatbot (Ashfaq et al., 2020). Tinwell (2009) indicated that anthropomorphic features of chatbots have positive effects on user satisfaction.

Earlier research explored users' perception of the chatbot (i.e., social presence, trust, and satisfaction) through the self-report method (Ahn et al., 2022; Jiang et al., 2022; Seitz et al., 2022). Informative as self-report measures may be, research also suggested that users suffer from certain disadvantages (Paulhus & Vazire, 2007). For example, earlier research showed that self-report measures can be biased by certain cognitive processes of the user, that is, people under certain circumstances report their inner states to be different from how they internally perceive them (Martin & Morich, 2011). Therefore, in the current research, we argue that a more objective measure (eye-tracking method) should be employed to reflect users' intrinsic perception. Users' eyes don't lie and eye movements will manifest users' visual behaviors and areas of attention (Guo et al., 2016). In addition, vision is a significant sense to capture users' perception of the interface design (Guo et al., 2022). Users' visual behaviors on the interface could be reflected by some eye-tracking metrics such as fixation count, fixation duration, and dwell time.

One of the most common manipulations of chatbots design is to use anthropomorphic appearances (Go & Sundar, 2019; Xu et al., 2022). Anthropomorphic appearances in chatbots are often designed with human pictures or animations to increase a feeling of humanness (Qiu & Benbasat, 2009). As the visual cue, anthropomorphic appearances could bring users a sense of sociality to the chatbot (Qiu & Benbasat, 2009). Evidence suggested that various types of appearances used in computer interfaces can have different kinds of effects on users' perception during the process of human-computer interaction (Nowak, 2004). Due to the finding that users generally respond more positively to chatbots with anthropomorphic characters (Chaves & Gerosa, 2021; Rhim et al., 2022), many studies have paid attention to the anthropomorphism of chatbots. That is, Araujo (2018) found that emotional connections between

user and chatbot can be enhanced after interacting with a chatbot that exhibits human-like cues. Further, female anthropomorphic chatbot design yielded more positive effects on humanness and warmth than male chatbot design (Borau et al., 2021; Pawlik, 2022). Also, Hassanein and Head (2007) claimed that chatbots using human pictures or animations as the appearance would increase perceived social presence of the chatbot. And likewise, Chattaraman et al. (2014) presented evidence that customers expressed more trust and satisfaction about websites with human-like appearances. However, Nowak (2004) found that low anthropomorphic appearances seem to be more likable and popular than high anthropomorphic appearances. A possible reason for this finding might be that users' perception of the appearance does not merely depend on the level of anthropomorphism, but rather that some other extrinsic factors and intrinsic cognition of users may also affect this relationship. Therefore, it seems that users' perception of the anthropomorphic appearances is complicated and diverse, and our perspective on this human perception can be enriched by not only measuring through the self-report method.

Apart from chatbot appearances, conversational styles are also a crucial cue used in chatbots design. As a special verbal cue in chatbot communication, conversational style refers to the linguistic variation utilized in the message that can induce different social responses (Feine et al., 2019). For example, Li and Wang (2023) suggested that a chatbot with informal (vs. formal) conversational style could increase continuance intention and improve brand attitude. Conversational styles of chatbots mainly include human-like styles and mechanical styles and the difference lies in the anthropomorphic degree of conversational language (Gnewuch et al., 2018). Existing research has shown that conversational styles of chatbots have effects on users' attitudes, personalization, and perception (Liebrecht & van Hooijdonk, 2020). Essentially, human-like conversational styles could evoke a human schema (Zhou et al., 2019), which in turn can affect users' perception and evaluation of the chatbot. Chatbots with human-like language styles (e.g., "May I help you?") displayed in the text dialogue can trigger the user to perceive the chatbot as an anthropomorphic agent and in turn improve users' perception of the chatbot (Liebrecht et al., 2021). Indeed, several studies showed that conversational styles have effects on users' perception. Specifically, Liebrecht et al. (2020) found that human-like conversational styles can yield a higher social presence and satisfaction with the chatbot. Araujo (2018) showed that users were more likely to interact with chatbots that used human-like conversational styles. Also, compared to mechanical styles, chatbots with human-like conversational styles can increase users' perceived trust and satisfaction (Muralidharan et al., 2014). Concludingly, chatbots with human-like conversational styles can enhance users' perception of social presence of the chatbot, trust in the chatbot, and satisfaction with the chatbot.

As shown by various (though separate) studies, both anthropomorphic appearances and conversational styles

have been found to affect users' perception of chatbots. Earlier research pointed out that high anthropomorphic characteristics may increase user expectations, which can result in disappointment when those high expectations are not satisfied (Mimoun et al., 2012). In addition, Baylor and Rosenberg-Kima (2006) argued that empathic text messages presented by the agent contribute to reducing frustration since people can attribute their frustration to technology rather than themselves.

However, to the best of our knowledge, although studied separately, earlier research has not investigated how these characteristics (appearances and conversational styles) affects users' perception together. It is conceivable that chatbots with anthropomorphic appearances and mechanical conversational styles may bring people less satisfaction (Egner, 2007). A feeling of incongruity may cause a less pleasant experience when chatbots partly show anthropomorphic features and partly do not. This incongruity also suggests an expectancy violation effect that one kind of high anthropomorphic cue may raise users' expectations for humanness to chatbots, but these expectations will be violated and even result in users less favorable attitudes and lower behavioral intentions when a chatbot also show another kind of low anthropomorphic cue (Go & Sundar, 2019). Additionally, in line with the literature on cognitive consistency principle, inconsistent cues (e.g., the pairing of a human face with a humanoid voice or a human voice with a humanoid face) will lead to longer cognitive processing time and less trust compared to consistent cues (the pairing of a human face with a human voice or a humanoid face with a humanoid voice) (Gong & Nass, 2007). Therefore, as two crucial features in the chatbot, we argue that the interaction effect between visual appearances and conversational styles on users' perception and attitudes to chatbots deserves to be investigated considering the wide usage of chatbots. Additionally, extant research primarily employed the self-reported method to measure user experience, which has the disadvantage of being fundamentally subjective and not clear about users' intrinsic perception. Hence, a more objective measurement method of user experiences seems essential to explore the interactive effects of these characters on users' perception.

1.2. Visual behaviors and eye-tracking metrics

In recent years, the eye-tracking technique has been widely used to reveal users' cognitive processes and behavioral intentions (Bialkova et al., 2020; Luan et al., 2016). As an objective measurement, eye tracking metrics provide useful data for the interface design (Wang et al., 2014), preference evaluation for humanoid robot appearance (Li, Guo, Fang, et al., 2022), and visual perception (Jarodzka et al., 2021). According to the eye-mind hypothesis (Just & Carpenter, 1976), people are thinking and perceiving the stimuli when they read sentences or view images, where the cognitive processes could be recognized by eye-tracking metrics. In addition, eye-tracking metrics also can be used to reveal users' perception and cognitive processes through

visual behaviors (Pei et al., 2022). When users interact with chatbots on the website, their visual behaviors to the anthropomorphic design cues of chatbots will be reflected in eye-tracking metrics.

The eye-tracking method obtains great popularity in exploring users' perception and attitudes. Wang et al. (2014) manifested that product images with a human face generate much appeal and social presence, which in turn leads to positive attitudes. Li, Guo, Ren, et al. (2022) showed that positive appearances of humanoid robots lead to short fixations than neutral or negative appearances, that is, more visual attention is allocated to neutral or negative appearances of humanoid robots. Boardman et al. (2022) indicated that rotating banners in the visual hierarchy of retail homepages can attract more users' attention and cognitive resources. Pei et al. (2022) argued that dynamic images are better than static images for users' experience. Several studies have applied an eye-tracking method to explore users' cognition and visual behaviors in different scenarios. However, no research has yet investigated users' perception and visual behaviors in the chatbot design, which directly causes the gap between users' visual perception and the chatbot design.

Among eye-tracking metrics, fixation count, fixation duration, and dwell time are the main eye-tracking metrics widely used to reflect users' perception and visual information. Fixation count refers to the number of fixations in the area of interest (AOI), which can explain the total amount of visual attention (Behe et al., 2015) or user preference (Glaholt et al., 2009). Average fixation duration is the average duration of each fixation (Holmqvist et al., 2011). The longer average fixation duration indicates more occupation of time in the area and deeper cognitive processing (Unema et al., 2005). In addition, longer average fixation duration may imply that the area is more attractive, or it represents more difficulties in information processing (Jacob & Karn, 2003). Dwell time is the sum time of fixation and saccade. Longer dwell time means higher perceived experience and attraction (Guo et al., 2016). This study attempted to use these eye-tracking metrics to analyze users' visual behaviors and explore the potential cognitive mechanism to anthropomorphic design cues on the chatbot.

1.3. Aim of this study

Briefly, anthropomorphic appearances and conversational styles of chatbots are two crucial features in human-chatbot interaction (Go & Sundar, 2019). Existing research separately examined anthropomorphic appearances or conversational styles of chatbots on users' attitudes and behavioral intentions with the self-reported method (Araujo, 2018; Roy & Naidoo, 2021; Xu et al., 2022). Additionally, little research has considered these two features together, and none has even used more objective measures, such as an eye-tracking method. Previous research contributing to exploring users' subjective perception of anthropomorphic appearances and conversational styles reaches limited consensus (Pawlik, 2022; Schuetzler et al., 2020). However, there still exists a critical research gap in exploring the joint effect on users'

perception and visual behaviors. Besides, a more objective eye-tracking method is necessary to reveal users' intrinsic perception of the chatbot. Grounded on the aforementioned findings, this study aimed to explore how anthropomorphic appearances and conversational styles of chatbots jointly affect users' perception and visual behaviors. The combined method of the subjective evaluation and an eye-tracking method was used to reflect users' perception and visual behaviors. Specifically, the following research questions were proposed:

RQ1: How do anthropomorphic appearances and conversational styles jointly affect users' perception of the chatbot?

RQ2: How do anthropomorphic appearances and conversational styles jointly affect users' visual behaviors?

2. Methodology

This study employed a 3 (appearance anthropomorphism: high anthropomorphic vs. medium anthropomorphic vs. low anthropomorphic) $\times$ 2 (conversational style: human-like vs. mechanical) within-subject experiment. An eye-tracking experiment was conducted to explore users' visual behaviors. Users' perception was collected with the subjective evaluation during the experiment.

2.1. Stimuli

The prototype of chatbots on the bank website was constructed for the experiment since chatbots often provide consulting services on many online bank websites (Lui & Lamb, 2018). The chatbot was designed as a regular customer service assistant in a fictitious bank website named XYZ, providing a short dialogue to user inquiries. To avoid the effect of the dialogue contents on the purpose of this experiment, the dialogue topic was limited to opening hours about the bank.

The chatbot was designed for six conditions based on the level of anthropomorphic appearances and conversational styles. Following earlier research (Appel et al., 2012; Araujo, 2018; Go & Sundar, 2019), females are more common used as the visual appearance in chatbot research. Additionally, earlier research showed that chatbot gender has significant impact on users' perception (Pawlik, 2022). Therefore, to avoid gender effects, high anthropomorphic appearances were designed with a real woman, and medium anthropomorphic appearances were designed with an animated woman, while low anthropomorphic appearances were designed with a "speech bubble" in the present study. Conversational styles were designed with the human-like style (e.g., hello, may I help you, and you are welcome) and the mechanical style (e.g., start and quit). To better measure users' perception and visual behaviors, 3 similar prototypes for each level of anthropomorphic appearances were designed. Hence, there are 18 different stimuli in this experiment (Appendix A). Additionally, these chatbot appearances were designed for females to avoid gender bias (Pawlik, 2022). An overview of the different prototypes of chatbots is shown in Table 1.

Table 1. Experiment prototypes.

Number	Anthropomorphic appearances	Conversational styles	Number	Anthropomorphic appearances	Conversational styles
1	High #1	Human-like	10	High #2	Mechanical
2	Medium #1	Human-like	11	Medium #2	Mechanical
3	Low #1	Human-like	12	Low #2	Mechanical
4	High #1	Mechanical	13	High #3	Human-like
5	Medium #1	Mechanical	14	Medium #3	Human-like
6	Low #1	Mechanical	15	Low #3	Human-like
7	High #2	Human-like	16	High #3	Mechanical
8	Medium #2	Human-like	17	Medium #3	Mechanical
9	Low #2	Human-like	18	Low #3	Mechanical

Before the formal experiment, a pre-study with 27 participants aged from 22 to 28 ($M=25.01$, $SD=1.49$) was conducted to check the validity of the manipulation. This pre-study used a 3 (appearance anthropomorphism: high anthropomorphic vs. medium anthropomorphic vs. low anthropomorphic) within-subject design (repeated measures design). For chatbot appearances manipulation check, the perceived anthropomorphism was evaluated with items on a 7-point Likert scale from Nowak and Rauh (2005) (e.g., “the chatbot looks very human,” “the chatbot looks very realistic”). Results of repeat-measures ANOVAs showed that there were significant differences [$F(2, 78) = 113.464$, $p < 0.001$] in different levels of anthropomorphic appearances with high ($M=5.07$, $SD=0.78$), medium ($M=4.05$, $SD=1.07$), and low levels ($M=1.56$, $SD=0.77$). Besides, the difference [$F(2, 78) = 0.566$, $p=0.570$] between three images in the high level of anthropomorphic appearances, the difference [$F(2, 78) = 0.095$, $p=0.910$] between three images in the medium level of anthropomorphic appearances and the difference [$F(2, 78) = 0.662$, $p=0.519$] between three images in the low level of anthropomorphic appearances were not significant. For conversational styles manipulation check, perceived conversational styles were evaluated with items on a 5-point Likert scale from Bartneck et al. (2009) (e.g., “my impression of the chatbot’s language felt humanlike,” “my impression of the chatbot’s language felt alive”). The results showed that there were significant differences [$t(52) = 6.899$, $p < 0.001$] in two different conversational styles with human-like style ($M=3.64$, $SD=1.07$) and mechanical style ($M=1.76$, $SD=0.93$). Therefore, manipulation checks for anthropomorphic appearances and conversational styles were valid.

2.2. Participants

A priori analysis using G-power 3.1 manifested that a minimal sample size of 72 would be required to detect a large effect size ($f=0.25$, $\alpha=0.05$, $1-\beta=0.90$) (Cohen, 1988). To obtain more valid data, a total of 80 university participants were recruited through campus posters. Data from 4 out of 80 participants were excluded due to the low eye-tracking accuracy. Therefore, there were 76 valid participants, including 36 males and 40 females. Participants’ age ranged from 21 to 30 ($M=24.33$, $SD=2.02$). Various majors were represented across the participant pool, including business, engineering, accounting, marketing, and finance. All participants were in good physical condition,

with normal or corrected-to-normal vision without astigmatism. After finishing the experiment, each participant would receive 20 RMB as compensation for voluntary participation. The study was approved by the Ethics Review Committee of the local university.

2.3. Instruments

The SMI-iView XTM RED eye-tracking system with an accuracy of 0.4° was used to record users’ visual movements at a sampling rate of 500 Hz. This system is composed of a recording computer, a 20-inch LCD monitor with a resolution of 1680×1050 pixels, and two remote infra-red diodes placed at the bottom. BeGaze 3.6 software was used to define each AOI and analyze eye-tracking metrics.

2.4. Procedure

The experiment was conducted in a human factor laboratory with sound insulation, reasonable lighting, and suitable temperature. Each participant was instructed about the experiment purposes, procedures, and signed an informed consent form voluntarily before the formal experiment. Additionally, participants were required to imagine the scenario that they ask the chatbot on the website of ABC bank about the opening hours of the workday. To collect quantitative and qualitative data, two main steps were included in the experiment. Before the experiment task, a calibration procedure was performed to control the error of eye position within 0.5° . In the first step, each participant freely viewed the chatbot without time limitations. In the second step, each participant was asked to complete a subjective evaluation to provide their intuitive perception of the chatbot. An LCD screen would automatically display the next chatbot after participants completed the evaluation in a self-determined time. To counter order effects, all stimuli of chatbots were presented in random throughout the experiment. The flow diagram of the eye-tracking experiment was shown in Figure 1. The whole experiment will not end until participants have viewed and evaluated all chatbots. For each participant, the total time of the procedure was approximately 25 min.

2.5. Measurements and eye-tracking metrics

In this study, we primarily observed users’ subjective perception and visual behaviors when freely viewing chatbots. The

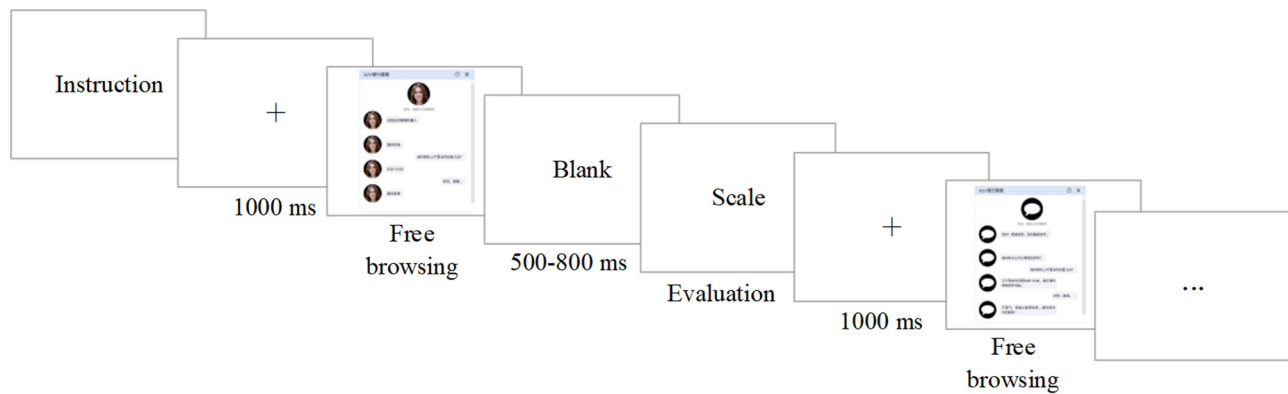


Figure 1. Flow diagram of eye-tracking experiment.



Figure 2. Definition of each AOI on the chatbot interface.

evaluation scale consisted of three constructs: perceived social presence, perceived trust, and satisfaction. All measurements were assessed on a 7-point Likert scale. Perceived social presence was measured with items from Gefen and Straub (2003). Perceived trust was measured with items from Lee and See (2004). Satisfaction was measured with items from Oliver (1980).

To analyze eye-tracking metrics, AOIs were defined for each trial. AOIs include the area of chatbot appearances, the area of dialogue contents, and the area of the whole chatbot interface. Figure 2 showed details about AOIs on the chatbot interface. To avoid the effect of AOI size on eye-tracking metrics, these metrics were standardized in the same size by dividing each AOI coverage (the percentage of AOI size in the whole size of stimuli) (Guo et al., 2010; Li, Guo, Ren, et al., 2022).

3. Results

The subjective evaluation and eye-tracking data were processed by within-subject repeated-measures ANOVAs, corrected with Greenhouse-Geisser correction by SPSS 25.0. The distribution will be transformed to a normal level if the raw data is skewed (Amin et al., 2019). Repeated-measures ANOVAs for eye-tracking metrics were conducted on the AOIs of the chatbot interface, chatbot appearances, and dialogue contents, respectively. An overview of the results was displayed in Table 2. This table shows that both main effects and interaction effects between the chatbot appearances and conversational styles on users' perception of social presence, users' trust, and users' satisfaction were found. While only some significant main effects but no interaction effect were found on eye-tracking metrics.

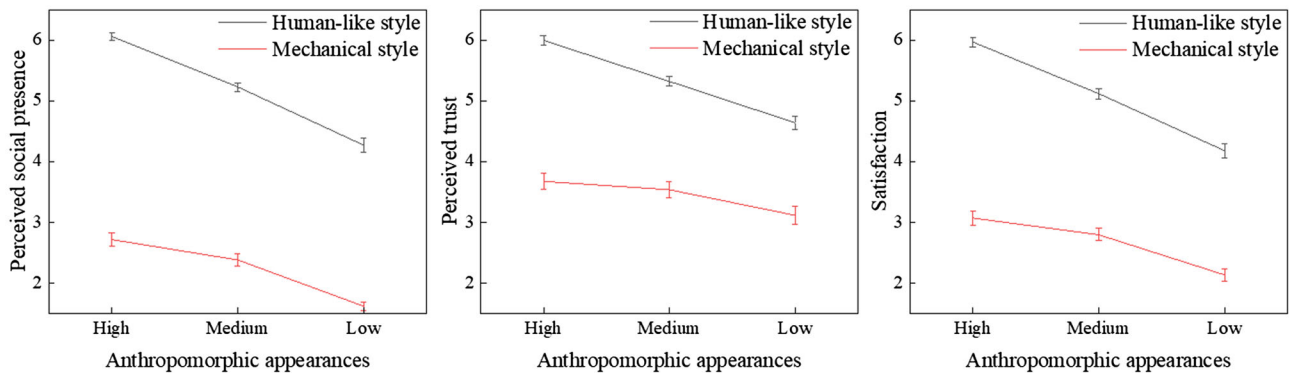
3.1. Subjective evaluation results

Repeat-measures ANOVAs were conducted on users' perception of social presence, users' trust, and users' satisfaction with anthropomorphic chatbot appearances (high vs. medium vs. low) and conversational styles (human-like vs. mechanical), respectively. The mean perceived social presence, perceived trust, and satisfaction (and their standard deviations) in different conditions were shown in Table 3.

The results of repeated-measures ANOVA indicated significant main effects of the chatbot appearances [$F(1.605, 121.143) = 214.619, p < 0.001, \eta_p^2 = 0.741$] and conversational styles [$F(1, 75) = 849.040, p < 0.001, \eta_p^2 = 0.919$] on perceived social presence. The interaction effect between the chatbot appearances and conversational styles was also found [$F(1.821, 136.574) = 16.841, p < 0.001, \eta_p^2 = 0.183$]. Simple effect analyses showed that in all conditions of chatbot appearances, perceived social presence scores of chatbot with human-like conversational styles were higher than that in chatbot with mechanical conversational styles. Similarly, in different kinds of conversational styles, perceived social presence scores were highest at high anthropomorphic appearances, followed by medium anthropomorphic appearances, and lowest at low anthropomorphic appearances, as shown in Figure 3. Furthermore, in the condition of high anthropomorphic appearance, users perceived a higher social

Table 2. An overview of the measured effects.

Measured outcomes		Effects	<i>F</i>	<i>p</i>	η_p^2
Subjective outcomes	Perceived social presence	Chatbot appearances	214.619	0.000	0.741
		Conversational styles	849.040	0.000	0.919
		Chatbot appearances $\times$ conversational styles	16.841	0.000	0.183
	Perceived trust	Chatbot appearances	73.012	0.000	0.493
		Conversational styles	243.636	0.000	0.765
		Chatbot appearances $\times$ conversational styles	22.744	0.000	0.233
	Satisfaction	Chatbot appearances	156.091	0.000	0.675
		Conversational styles	513.608	0.000	0.873
		Chatbot appearances $\times$ conversational styles	30.413	0.000	0.289
Eye-tracking outcomes	Fixation count on the area of the whole chatbot interface	Chatbot appearances	7.423	0.001	0.090
		Conversational styles	72.494	0.000	0.492
		Chatbot appearances $\times$ conversational styles	0.350	0.705	0.005
	Average fixation duration on the area of the whole chatbot interface	Chatbot appearances	0.887	0.414	0.012
		Conversational styles	27.561	0.000	0.269
		Chatbot appearances $\times$ conversational styles	0.089	0.915	0.001
	Dwell time on the area of the whole chatbot interface	Chatbot appearances	10.981	0.000	0.128
		Conversational styles	74.539	0.000	0.498
		Chatbot appearances $\times$ conversational styles	0.185	0.831	0.002
	Fixation count on the area of chatbot appearance	Chatbot appearances	56.436	0.000	0.429
		Conversational styles	0.609	0.437	0.008
		Chatbot appearances $\times$ conversational styles	0.126	0.882	0.002
	Average fixation duration on the area of chatbot appearance	Chatbot appearances	42.867	0.000	0.364
		Conversational styles	2.845	0.096	0.037
		Chatbot appearances $\times$ conversational styles	0.891	0.412	0.012
	Dwell time on the area of chatbot appearance	Chatbot appearances	52.544	0.000	0.412
		Conversational styles	4.055	0.084	0.051
		Chatbot appearances $\times$ conversational styles	0.485	0.616	0.006
	Fixation count on the area of dialogue contents	Chatbot appearances	3.573	0.030	0.045
		Conversational styles	89.647	0.000	0.544
		Chatbot appearances $\times$ conversational styles	0.275	0.760	0.004
	Average fixation duration on the area of dialogue contents	Chatbot appearances	4.469	0.027	0.056
		Conversational styles	60.940	0.000	0.448
		Chatbot appearances $\times$ conversational styles	0.110	0.798	0.001
	Dwell time on the area of dialogue contents	Chatbot appearances	7.852	0.001	0.095
		Conversational styles	85.098	0.000	0.532
		Chatbot appearances $\times$ conversational styles	0.514	0.583	0.007

**Figure 3.** Interaction effects between anthropomorphic appearances and conversational styles on perceived social presence, perceived trust, and satisfaction.

presence to chatbots with a human-like conversational style compared with a mechanical conversational style, and this difference was larger than when the chatbot had a medium or low anthropomorphic appearance. Post-hoc analyses also showed that users perceived social presence scores to chatbots with human-like conversational styles were higher than that chatbots with mechanical conversational styles in conditions of high anthropomorphic appearances ($p < 0.001$), medium anthropomorphic appearances ($p < 0.001$) and low anthropomorphic appearances ($p < 0.001$). Likewise, users perceived social presence scores were highest at high anthropomorphic appearances, followed by medium anthropomorphic appearances, and lowest at low

anthropomorphic appearances in conditions of human-like conversational styles ($ps < 0.001$) and mechanical conversational styles ($ps < 0.001$).

The results of repeated-measures ANOVA indicated significant main effects of the chatbot appearances [$F(1.605, 120.350) = 73.012$, $p < 0.001$, $\eta_p^2 = 0.493$] and conversational styles [$F(1, 75) = 243.636$, $p < 0.001$, $\eta_p^2 = 0.765$] on perceived trust. The interaction effect between the chatbot appearances and conversational styles was also found [$F(1.657, 124.252) = 22.744$, $p < 0.001$, $\eta_p^2 = 0.233$]. Simple effect analyses showed that in all conditions of chatbot appearances, perceived trust scores of chatbot with human-like conversational styles were higher than that in chatbot

with mechanical conversational styles. Similarly, in different kinds of conversational styles, perceived trust scores were highest at high anthropomorphic appearances, followed by medium anthropomorphic appearances, and lowest at low anthropomorphic appearances, as shown in Figure 3. In addition, in the condition of high anthropomorphic appearance, users perceived trust in chatbots with a human-like conversational style were higher compared with a mechanical conversational style, and this difference was larger than when the chatbot had a medium or low anthropomorphic appearance. Post-hoc analyses also showed that users perceived trust scores to chatbots with human-like conversational styles were higher than that chatbots with mechanical conversational styles in conditions of high anthropomorphic appearances ($p < 0.001$), medium anthropomorphic appearances ($p < 0.001$) and low anthropomorphic appearances ($p < 0.001$). Likewise, users perceived trust scores were highest at high anthropomorphic appearances, followed by medium anthropomorphic appearances, and lowest at low anthropomorphic appearances in conditions of human-like conversational styles ($ps < 0.001$) and mechanical conversational styles [high-medium ($p = 0.101$), high-low ($p < 0.001$) and medium-low ($p < 0.001$)].

The results of repeated-measures ANOVA manifested significant main effects of the chatbot appearances [$F(1.622, 121.658) = 156.091, p < 0.001, \eta_p^2 = 0.675$] and conversational styles [$F(1, 75) = 513.608, p < 0.001, \eta_p^2 = 0.873$] on user satisfaction. The interaction effect between the chatbot appearances and conversational styles was also found [$F(2, 150) = 30.413, p < 0.001, \eta_p^2 = 0.289$]. Simple effect analyses showed that in all conditions of chatbot appearances, user satisfaction scores of chatbot with human-like conversational styles were higher than that in chatbot with mechanical conversational styles. Similarly, in different kinds of conversational styles, users' satisfaction scores were highest at high anthropomorphic appearances, followed by medium anthropomorphic appearances, and lowest at low anthropomorphic appearances, as shown in Figure 3. What's more, in the condition of high anthropomorphic appearance, user satisfaction with chatbots with a human-like conversational style were higher compared with a mechanical conversational style, and this difference was larger than when the chatbot had a medium or low anthropomorphic appearance. Post-hoc analyses also showed that user satisfaction scores to chatbots with human-like conversational styles were higher than that chatbots with mechanical conversational styles in conditions of high anthropomorphic appearances ($p < 0.001$), medium anthropomorphic appearances ($p < 0.001$) and low anthropomorphic appearances

($p < 0.001$). Likewise, user satisfaction scores were highest at high anthropomorphic appearances, followed by medium anthropomorphic appearances, and lowest at low anthropomorphic appearances in conditions of human-like conversational styles ($ps < 0.001$) and mechanical conversational styles ($ps < 0.001$).

3.2. Eye-tracking outcomes

3.2.1. Eye-tracking metrics on the area of the whole chatbot interface

Table 4 displayed the descriptive statistics of eye-tracking metrics on the area of the whole chatbot interface. The results of repeated-measures ANOVA found significant main effects of the chatbot appearances [$F(2, 150) = 7.423, p < 0.001, \eta_p^2 = 0.090$] and conversational styles [$F(1, 75) = 72.494, p < 0.001, \eta_p^2 = 0.492$] on fixation count. No interaction effect was observed. More specific trends were denoted by multiple comparisons of fixation count, as shown in Figure 4. The results revealed that chatbots with high anthropomorphic appearances ($M = 0.764, SD = 0.045$) attracted more fixation counts than chatbots with low anthropomorphic appearances ($M = 0.691, SD = 0.037, p < 0.01$), and chatbots with medium anthropomorphic appearances ($M = 0.760, SD = 0.041$) attracted more fixation counts than chatbots with low anthropomorphic appearances ($M = 0.691, SD = 0.037, p < 0.01$). The difference between chatbots with high and medium anthropomorphic appearances was not significant ($p = 0.833$). As for conversational styles, the result showed that chatbots with human-like styles ($M = 0.832, SD = 0.046$) attracted more fixation counts than chatbots with mechanical styles ($M = 0.645, SD = 0.034, p < 0.001$).

The results of repeated-measures ANOVA showed a significant main effect of conversational styles [$F(1, 75) = 27.561, p < 0.001, \eta_p^2 = 0.269$] on average fixation duration, while no significant main effect on the chatbot appearances and no interaction effect were found. More specific trends were denoted by multiple comparisons of average fixation duration, as shown in Figure 4. The results revealed that chatbots with mechanical styles attracted longer average fixation duration ($M = 3.477, SD = 0.124$) than chatbots with human-like styles ($M = 3.261, SD = 0.099, p < 0.001$).

The results of repeated-measures ANOVA manifested significant main effects of the chatbot appearances [$F(2, 150) = 10.981, p < 0.001, \eta_p^2 = 0.128$] and conversational styles [$F(1, 75) = 74.539, p < 0.001, \eta_p^2 = 0.498$] on dwell time. No interaction effect was observed. More specific trends

Table 3. Descriptive statistics of perceived social presence, perceived trust, and satisfaction.

Condition		Perceived social presence	Perceived trust	Satisfaction
Appearances anthropomorphism	Conversational Style	<i>M</i> (<i>SD</i>)	<i>M</i> (<i>SD</i>)	<i>M</i> (<i>SD</i>)
High	Human-like	6.057 (0.580)	5.941 (0.667)	5.965 (0.685)
High	Mechanical	2.713 (0.959)	3.673 (1.161)	3.068 (0.973)
Medium	Human-like	5.226 (0.618)	5.318 (0.680)	5.116 (0.755)
Medium	Mechanical	2.380 (0.893)	3.535 (1.147)	2.794 (0.893)
Low	Human-like	4.270 (1.041)	4.636 (0.989)	4.178 (1.038)
Low	Mechanical	1.617 (0.636)	3.114 (1.301)	2.132 (0.895)

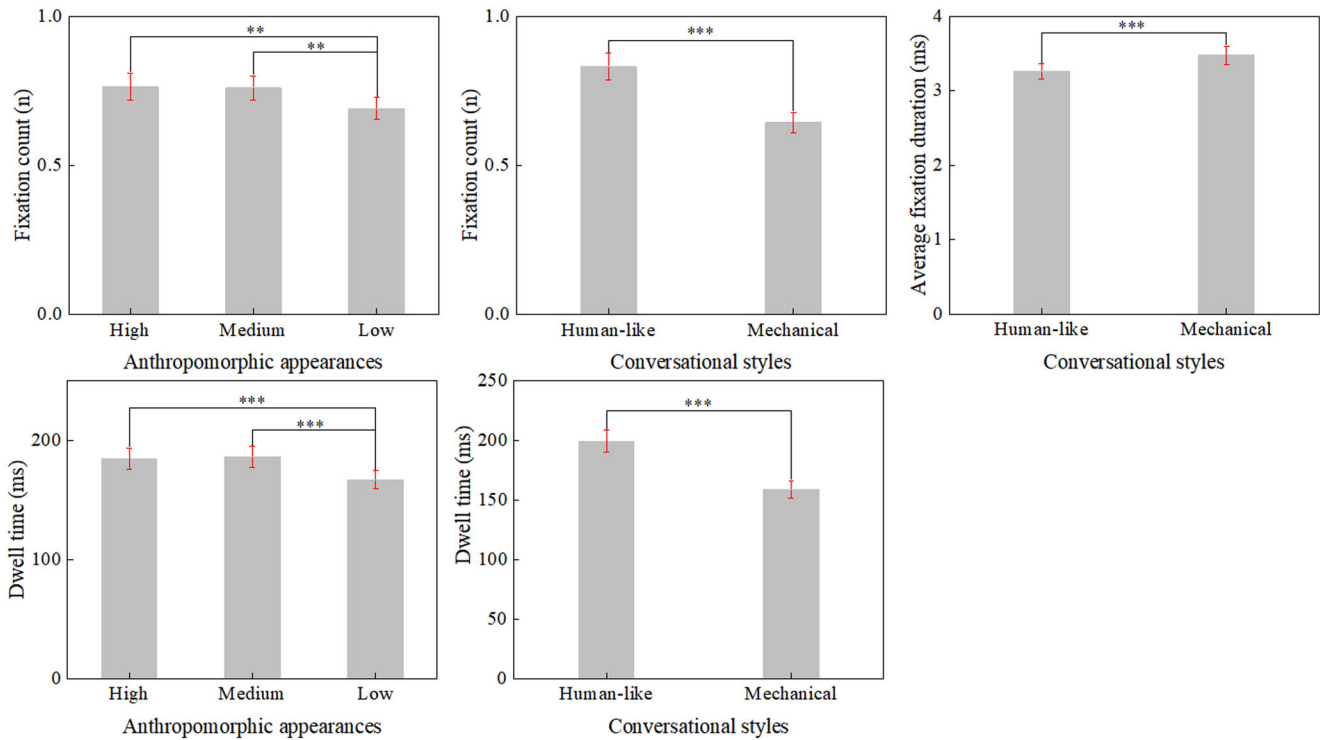


Figure 4. Multiple comparisons of eye-tracking metrics on the area of the whole chatbot interface.

Table 4. Descriptive statistics of eye-tracking metrics on the area of the whole chatbot interface.

Condition		Fixation count (n)	Average fixation duration (ms)	Dwell time (ms)
Appearances anthropomorphism	Conversational Style	M (SD)	M (SD)	M (SD)
High	Human-like	0.860 (0.465)	3.290 (0.895)	203.952 (89.545)
High	Mechanical	0.668 (0.362)	3.501 (1.159)	164.725 (79.384)
Medium	Human-like	0.860 (0.434)	3.251 (0.920)	205.071 (91.779)
Medium	Mechanical	0.659 (0.313)	3.485 (1.167)	166.578 (71.766)
Low	Human-like	0.776 (0.393)	3.241 (0.888)	188.704 (79.77)
Low	Mechanical	0.607 (0.317)	3.445 (1.023)	145.041 (65.449)

were denoted by multiple comparisons of dwell time, as shown in Figure 4. The results revealed that dwell time in chatbot with high anthropomorphic appearances ($M = 184.338$, $SD = 9.041$) was longer than that in chatbot with low anthropomorphic appearances ($M = 166.873$, $SD = 7.578$, $p < 0.001$), and dwell time in chatbot with medium anthropomorphic appearances ($M = 185.824$, $SD = 8.763$) was also longer than that in chatbot with low anthropomorphic appearances ($M = 166.873$, $SD = 7.578$, $p < 0.001$). The difference between chatbot with high and medium anthropomorphic appearances was not significant ($p = 0.728$). As for conversational styles, the result showed that dwell time in the chatbot with human-like style ($M = 199.242$, $SD = 9.315$) was longer than that in the chatbot with mechanical style ($M = 158.781$, $SD = 7.395$, $p < 0.001$).

3.2.2. Eye-tracking metrics on the area of chatbot appearances

Table 5 displayed the descriptive statistics of eye-tracking metrics on the area of chatbot appearances. The results of repeated-measures ANOVA showed a significant main effect

of the chatbot appearances [$F(1.548, 116.097) = 56.436$, $p < 0.001$, $\eta_p^2 = 0.429$] on fixation count, while no significant main effect on the conversational style and no interaction effect were found. More specific trends were denoted by multiple comparisons of fixation count, as shown in Figure 5. The results revealed that chatbots with high anthropomorphic appearances ($M = 2.923$, $SD = 0.226$) attracted more fixation counts than chatbots with low anthropomorphic appearances ($M = 1.646$, $SD = 0.165$, $p < 0.001$), and chatbots with medium anthropomorphic appearances ($M = 2.784$, $SD = 0.217$) attracted more fixation counts than chatbots with low anthropomorphic appearances ($M = 1.646$, $SD = 0.165$, $p < 0.001$). However, the difference between chatbots with high anthropomorphic appearances and medium anthropomorphic appearances was not significant ($p = 0.185$).

The results of repeated-measures ANOVA showed a significant main effect of the chatbot appearances [$F(2, 150) = 42.867$, $p < 0.001$, $\eta_p^2 = 0.364$] on average fixation duration, while no significant main effect on conversational styles and no interaction effect were found. More specific trends were denoted by multiple comparisons of average fixation duration, as shown in Figure 5. The results manifested that

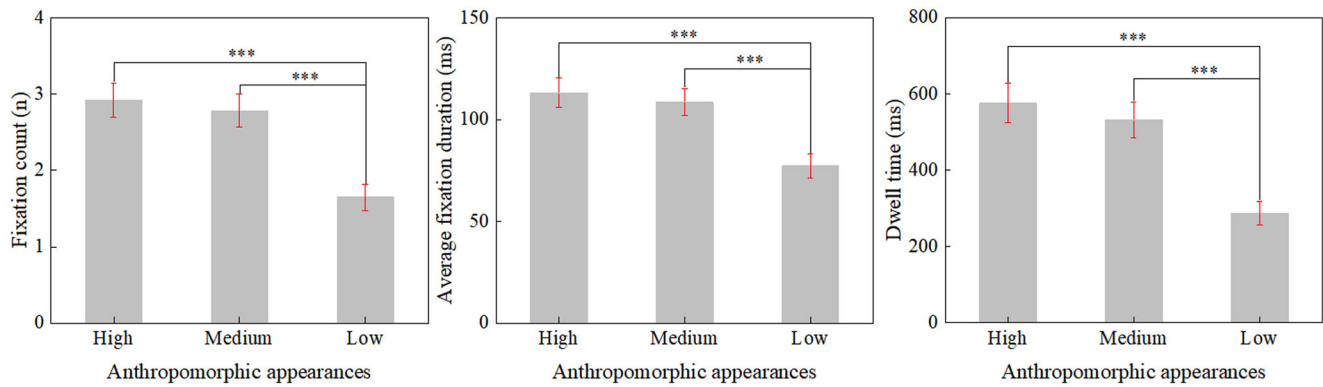


Figure 5. Multiple comparisons of eye-tracking metrics in the area of chatbot appearances.

Table 5. Descriptive statistics of eye-tracking metrics on the area of chatbot appearances.

Condition		Fixation count (n)	Average fixation duration (ms)	Dwell time (ms)
Appearances anthropomorphism	Conversational Style	<i>M</i> (<i>SD</i>)	<i>M</i> (<i>SD</i>)	<i>M</i> (<i>SD</i>)
High	Human-like	2.905 (2.029)	112.488 (68.439)	565.544 (446.770)
High	Mechanical	2.942 (2.179)	113.718 (70.968)	585.045 (504.533)
Medium	Human-like	2.753 (2.079)	105.735 (62.154)	500.677 (390.251)
Medium	Mechanical	2.814 (1.982)	111.556 (67.417)	562.808 (484.403)
Low	Human-like	1.578 (1.537)	71.113 (57.296)	266.553 (270.829)
Low	Mechanical	1.713 (1.520)	83.207 (52.751)	309.654 (293.393)

chatbots with high anthropomorphic appearances ($M=113.103$, $SD=7.247$) attracted longer average fixation duration than chatbots with low anthropomorphic appearances ($M=77.160$, $SD=5.782$, $p<0.001$), and chatbots with medium anthropomorphic ($M=108.646$, $SD=6.768$) appearances attracted longer average fixation duration than chatbots with low anthropomorphic appearances ($M=77.160$, $SD=5.782$, $p<0.001$). However, the difference between chatbots with high anthropomorphic appearances and medium anthropomorphic appearances was not significant ($p=0.247$).

The results of repeated-measures ANOVA manifested significant main effects of the chatbot appearances [$F(1.461, 109.612) = 52.544$, $p<0.001$, $\eta_p^2 = 0.412$] on dwell time, while no significant main effect on conversational styles and no interaction effect were found. More specific trends were denoted by multiple comparisons of dwell time, as shown in Figure 5. The results revealed that dwell time in chatbots with high anthropomorphic appearances ($M=575.295$, $SD=51.357$) was longer than that in chatbots with low anthropomorphic appearances ($M=288.103$, $SD=30.405$, $p<0.001$), and dwell time in chatbots with medium anthropomorphic appearances ($M=531.742$, $SD=47.233$) was also longer than that in chatbots with low anthropomorphic appearances ($M=288.103$, $SD=30.405$, $p<0.001$). However, the difference between chatbots with high anthropomorphic appearances and medium anthropomorphic appearances was not significant ($p=0.051$).

3.2.3. Eye-tracking metrics on the area of dialogue contents

Table 6 displayed the descriptive statistics of eye-tracking metrics on the area of conversation. The results of repeated-

measures ANOVA found significant main effects of the chatbot appearances [$F(2, 150) = 3.573$, $p<0.05$, $\eta_p^2 = 0.045$] and conversational styles [$F(1, 75) = 89.647$, $p<0.001$, $\eta_p^2 = 0.544$] on fixation count. No interaction effect was observed. More specific trends were denoted by multiple comparisons of fixation count, as shown in Figure 6. The results revealed that chatbots with low anthropomorphic appearances ($M=1.120$, $SD=0.066$) attracted more fixation counts than chatbots with high anthropomorphic appearances ($M=1.032$, $SD=0.065$, $p<0.05$), and chatbots with medium anthropomorphic appearances ($M=1.102$, $SD=0.067$) attracted more fixation counts than chatbots with high anthropomorphic appearances ($M=1.032$, $SD=0.065$, $p<0.05$). However, the difference between chatbots with medium anthropomorphic appearances and low anthropomorphic appearances was not significant ($p=0.579$). As for conversational styles, the result showed that the chatbot with human-like style ($M=1.256$, $SD=0.076$) attracted more fixation counts than the chatbot with mechanical style ($M=0.913$, $SD=0.053$, $p<0.001$).

The results of repeated-measures ANOVA showed significant main effects of the chatbot appearances [$F(1.311, 98.330) = 4.469$, $p<0.05$, $\eta_p^2 = 0.056$] and conversational styles [$F(1, 75) = 60.940$, $p<0.001$, $\eta_p^2 = 0.448$] on average fixation duration, while there was no interaction effect. More specific trends were denoted by multiple comparisons of average fixation duration, as shown in Figure 6. The results revealed that chatbots with low anthropomorphic appearances ($M=7.473$, $SD=0.188$) attracted longer average fixation duration than chatbots with high anthropomorphic appearances ($M=6.991$, $SD=0.236$, $p=0.024$), and chatbots with medium anthropomorphic appearances attracted longer average fixation duration ($M=7.471$, $SD=0.254$) than chatbots with high anthropomorphic appearances ($M=6.991$,

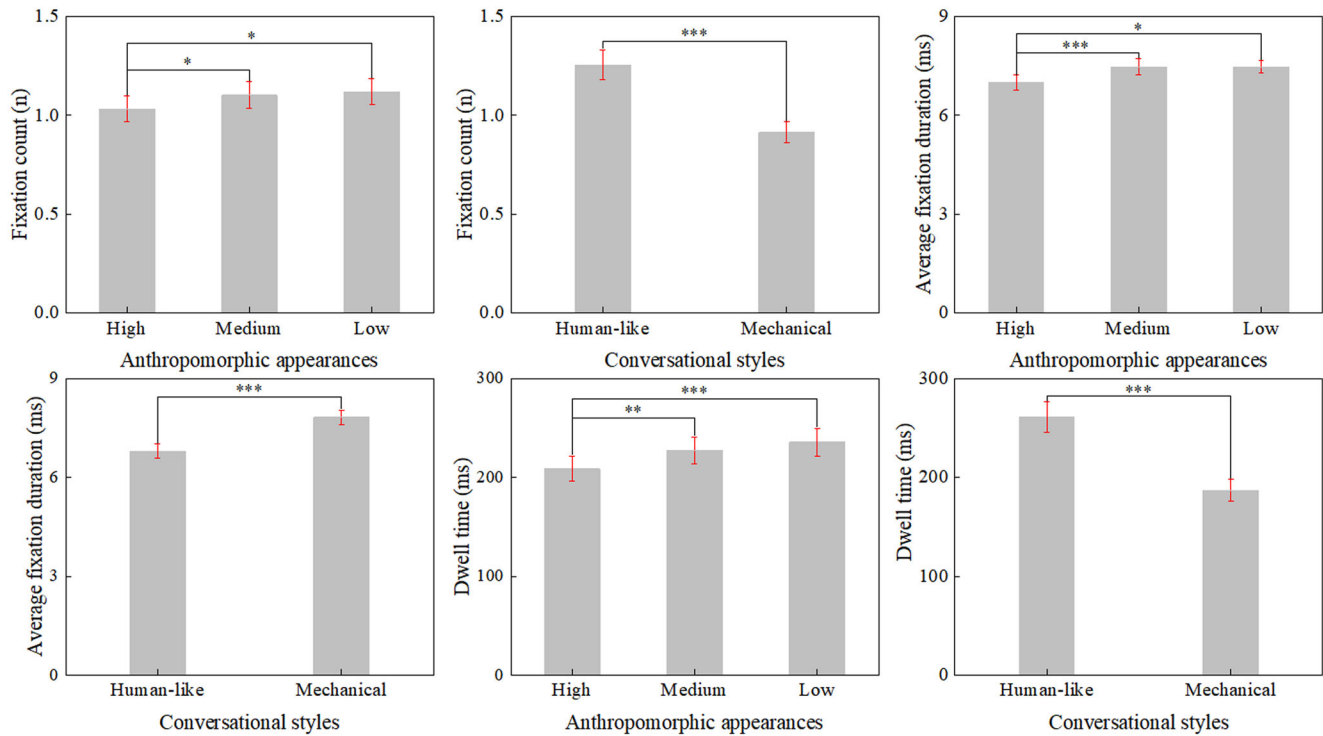


Figure 6. Multiple comparisons of eye-tracking metrics on the area of dialogue contents.

Table 6. Descriptive statistics of eye-tracking metrics on the area of dialogue contents.

Condition		Fixation count (n)	Average fixation duration (ms)	Dwell time (ms)
Appearances anthropomorphism	Conversational Style	<i>M</i> (<i>SD</i>)	<i>M</i> (<i>SD</i>)	<i>M</i> (<i>SD</i>)
High	Human-like	1.193 (0.707)	6.525 (1.819)	241.739 (134.354)
High	Mechanical	0.871 (0.498)	7.457 (2.447)	175.145 (102.783)
Medium	Human-like	1.287 (0.727)	6.926 (1.959)	264.566 (151.119)
Medium	Mechanical	0.916 (0.511)	8.016 (2.652)	189.651 (107.566)
Low	Human-like	1.288 (0.706)	6.936 (1.904)	276.053 (148.253)
Low	Mechanical	0.951 (0.537)	8.009 (2.662)	195.256 (114.037)

$SD = 0.236$, $p < 0.001$). However, the difference between chatbots with medium anthropomorphic appearances and low anthropomorphic appearances was not significant ($p = 0.994$). As for conversational styles, the result showed that the chatbot with mechanical style ($M = 7.827$, $SD = 0.216$) attracted longer average fixation duration than the chatbot with human-like style ($M = 6.796$, $SD = 0.207$, $p < 0.001$).

The results of repeated-measures ANOVA manifested significant main effects of the chatbot appearances [$F(2, 150) = 7.852$, $p < 0.005$, $\eta_p^2 = 0.095$] and conversational styles [$F(1, 75) = 85.098$, $p < 0.001$, $\eta_p^2 = 0.532$] on dwell time. No interaction effect was observed. More specific trends were denoted by multiple comparisons of dwell time, as shown in Figure 6. The results revealed that dwell time in chatbots with low anthropomorphic appearances ($M = 235.654$, $SD = 13.914$) was longer than that in chatbots with high anthropomorphic appearances ($M = 208.442$, $SD = 12.633$, $p < 0.001$), and dwell time in chatbots with medium anthropomorphic appearances ($M = 227.109$, $SD = 13.915$) was also longer than that in chatbots with high anthropomorphic appearances ($M = 208.442$, $SD = 12.633$, $p < 0.01$). However, the difference between chatbots with

medium anthropomorphic appearances and low anthropomorphic appearances was not significant ($p = 0.210$). As for conversational styles, the result showed that dwell time in the chatbot with human-like style ($M = 260.786$, $SD = 15.469$) was longer than that in the chatbot with mechanical style ($M = 186.684$, $SD = 11.164$, $p < 0.001$).

4. Discussion

With the aim of exploring the interaction effect of anthropomorphic appearances and conversational styles on users' subjective perception and visual behaviors of the chatbot, this study designed chatbots with three different levels of anthropomorphic appearances (high, medium, and low) and two kinds of conversational styles (human-like style and mechanical style). The eye-tracking method and a subjective evaluation were employed to record users' eye-tracking metrics and subjective perception. The results showed that anthropomorphic appearances and conversational styles had joint and diverse effects on users' perception and visual behaviors.

The results indicated that anthropomorphic appearances and human-like conversational styles jointly increased users'

perception of the chatbot's social presence, trust in the chatbot, and satisfaction with the chatbot. In contrast, on users' visual behaviors, such a joint effect was not found, although chatbots with higher anthropomorphic appearances and chatbots with human-like conversational styles triggered more fixation counts and longer dwell time. These findings suggested that anthropomorphic appearance and human-like conversational style can improve users' perception and attract more visual attention to the chatbot.

4.1. Effects on users' perception

Furthermore, the present results indicated that chatbot appearances and conversational styles had significant main effects on users' perception of social presence, users' trust, and user satisfaction. It is noteworthy that interaction effects between chatbot appearances and conversational styles on users' perception of social presence, users' trust, and user satisfaction were also found, which implied the joint effects of anthropomorphic appearances and conversational styles on users' perception. Specifically, regardless of the anthropomorphic level of chatbot appearances, users' perception of social presence, users' trust, and user satisfaction with human-like conversational styles were higher than that of mechanical conversational styles. Besides, under different conversational styles, users' perception of social presence, users' trust, and user satisfaction were positively related to the anthropomorphic level of chatbot appearances.

These results are in line with some earlier findings. Based on the research from Qiu and Benbasat (2009), users' perception of social presence of the chatbot with a human appearance was the highest, followed by chatbots with animated appearances, and the company logo with the lowest perceived social presence. Araujo (2018) and Liebrecht et al. (2021) also indicated that human-like language styles positively affected users' perception of social presence. As for the effect on perceived trust, Følstad et al. (2018) found that anthropomorphic appearances and conversational styles could increase users' perception of trust. Muralidharan et al. (2014) manifested that chatbots with human-like conversational styles have higher ratings of trust compared to mechanical styles. As regards satisfaction, Tinwell (2009) revealed that anthropomorphic appearances have positive effects on user satisfaction. Araujo (2018) also proved that users are more satisfied with chatbots with human-like language styles. In this study, anthropomorphic design cues yielded higher perception of social presence, users' trust, and user satisfaction. Additionally, according to the theory of social presence, higher perceived social presence could also promote users' trust and satisfaction (Cyr et al., 2007; Lu et al., 2016; Verhagen et al., 2014).

More importantly, the results manifested joint effects of these anthropomorphic design cues on users' subjective perception. We argue that finding an interaction between anthropomorphic appearance and human-like conversational style on subjectively measures only suggested that it is a higher-level cognitive effect that occurs: A chatbot that appears as highly anthropomorphic may set the expectation

that it interacts at high levels of anthropomorphism in all its utterances and behaviors. But, as our results showed, when chatbots with high anthropomorphic appearance but communicating in a mechanical language style, this kind of chatbot does not fit the user's expectations, and its social cues (appearance and language) are not aligned. For chatbots with medium or low levels of anthropomorphism in appearance, users may not have too high expectations of anthropomorphism, and failure of communicating in a humanlike style (but rather using a mechanical style) seem to have a less severe effect in the users' perceived social presence, perceived trust, and satisfaction. Also, these findings can be explained by the idea that users feel more natural and pleasant when chatbots with anthropomorphic appearances show conversational styles, which is consistent with their identity (as indicated by the type of conversational styles they use). This result was also similar to the find from Go and Sundar (2019) that there was an interaction effect between visual cues and identity cues of chatbots on perceived homophily. When the chatbot had anthropomorphic appearances, users in the human identity condition manifested higher perceived homophily and favorable attitudes. However, in the condition of mechanical conversational style, users' perception of social presence, users' trust, and user satisfaction with chatbots with different anthropomorphic appearances were especially low, even the chatbot with high anthropomorphic appearances. This finding confirmed the expectation violation effect (Burgoon & Walther, 1990). Chatbots with mechanical language styles made users expect machine-like cues, but the use of high anthropomorphic appearances may be perceived as dishonest and thereby reduced users' perception and behavioral attitudes. Another possible support for these joint effects could be offered by the media equation theory (Reeves & Nass, 1996), people instinctively attribute human traits to computers and treat them as social actors. Therefore, the superposition of two different anthropomorphic cues increased users' perception of social presence, users' trust, and user satisfaction. Different from existing research (Ciechanowski et al., 2019), this study did not find the uncanny valley effect. This might be because the chatbot appearances occupied a relatively small proportion of the area on the interface. Users were required to not only pay attention to the chatbot appearance but also follow conversational styles.

4.2. Effects on users' visual behaviors

Eye-tracking metrics revealed interesting findings. The present study observed significant main effects of chatbot appearances and conversational styles on users' fixation count and dwell time on the AOIs of the whole chatbot interface. Fixation count is typically associated with users' attention allocated to AOIs and may reflect the attractiveness of stimuli (Behe et al., 2015). Dwell time is the sum time of fixation and saccade and is related to user experience and interest (Guo et al., 2016). In the present study, users devoted more fixation counts and dwell time to chatbots with high and medium anthropomorphic appearances

compared to chatbots with low anthropomorphic appearances. Besides, chatbots with human-like conversational styles attracted more fixation counts and longer dwell time than chatbots with mechanical styles. These findings suggested that anthropomorphic design cues might attract users' interest and result in more visual attention (Araujo, 2018). Existing research manifested that using human images or animations could increase user satisfaction and lead to favorable attitudes (Hassanein & Head, 2007). Therefore, users devoted more visual attention to these anthropomorphic features. Moreover, the result showed a significant main effect of conversational styles on average fixation duration on the AOIs of the whole chatbot interface. Specifically, chatbots with mechanical conversational styles attracted longer average fixation duration than chatbots with human-like styles. The average fixation duration is related to the total amount of visual attention paid to each fixation and is considered attractive or a reflection of the processing depth of visual information (Ramos Gameiro et al., 2017; Unema et al., 2005). This finding indicated that mechanical language styles require deep information processing and further understanding. According to the media equation theory (Reeves & Nass, 1996), users instinctively react to artificial agents in the same way as to other people. When chatbots showed a mechanical language style, users may hesitate and need further confirmation of the information, thereby increasing the average fixation duration.

When separately investigating users' visual behaviors on the AOIs of chatbot appearances, results manifested significant main effects of chatbot appearances on users' fixation count, average fixation duration, and dwell time. Specifically, users devoted more fixation counts, longer average fixation duration, and longer dwell time to chatbots with high and medium anthropomorphic appearances compared to chatbots with low anthropomorphic appearances. The findings revealed that anthropomorphic design features attracted users' more visual attention. It also meant users were more interested in anthropomorphic appearances of chatbots, thereby causing sustained attention. This result is partially supported by the research about product appearances from Qu and Guo (2019) that pleasant appearances could yield the most visual attention. In the present study, anthropomorphic design features have been found to increase users' satisfaction and pleasantness (Go & Sundar, 2019; Nowak & Rauh, 2005). Whereas, present findings are different from the result that more visual attention are allocated to the neutral or negative appearance of humanoid robots (Li, Guo, Ren, et al., 2022). The possible reason is that neutral or negative appearances of humanoid robots lead to the uncanny valley effect (Ciechanowski et al., 2019) and give birth to negative or fearful impressions, which may activate negative biases and attract more visual attention. By contrast, the area of chatbot appearances only accounts for a small part of the whole interface. Therefore, the present study didn't show the uncanny valley effect. In addition, no significant difference between high and medium anthropomorphic appearances was confirmed. This probably because both human images and animated images can make users

feel anthropomorphic features, compared to the "speech bubble," which leads to more visual attention to the human images and animated images. Hence, although findings of the pre-study showed significant differences between different levels of anthropomorphic appearances, no significant difference between high and medium anthropomorphic appearances for eye-tracking metrics was found.

Regarding the AOIs of the dialogue contents, results indicated significant main effects of chatbot appearances and conversational styles on users' fixation count, average fixation duration, and dwell time. Specifically, chatbots with low anthropomorphic appearances attracted the most fixation count, the longest average fixation duration, and the longest dwell time on the AOIs of the dialogue contents, followed by chatbots with medium anthropomorphic appearances, and then chatbots with high anthropomorphic appearances. That is, users devoted more attention to high anthropomorphic appearances of chatbots, which caused less attention allocation to AOIs of the dialogue contents. This finding also suggested that anthropomorphic appearances are conducive to attracting users' visual attention. Earlier research showed that anthropomorphic design cues could adapt user preferences and provide a good first impression, thereby increasing users' interest and attention (Cao, Zhang, et al., 2021; Ho & Hou, 2015). As for the condition of conversational style, results indicated that chatbots with human-like styles attracted more fixation counts and longer dwell time than chatbots with mechanical styles. These findings also suggested that anthropomorphic styles might attract users' interest and result in more attention (Araujo, 2018). However, users devoted longer average fixation duration to the chatbot with mechanical conversational style rather than the chatbot with human-like style. This result provided additional support for the media equation theory (Reeves & Nass, 1996). Users treat interactions with chatbots the same way they view interactions with humans. When chatbots showed mechanical language styles rather than human-like styles, users may get frustrated and require more fixation duration to process information (Lee, 2008).

Nevertheless, this study did not observe any interaction effects of anthropomorphic appearances and conversational styles on eye-tracking metrics. This result might because significant effects of anthropomorphic appearances on eye-tracking metrics on the AOIs of the dialogue contents were confirmed, while there was no significant effect of conversational styles on eye-tracking metrics on the AOIs of the chatbot appearances. Overall, the result of eye-tracking metrics might suggest that anthropomorphic design cues may attract users' interest and result in more visual attention. This also can be supported by results from users' subjective perception that anthropomorphic appearances and human-like conversational styles could enhance users' perception of social presence, users' trust, and user satisfaction (Araujo, 2018; Følstad et al., 2018; Liebrecht et al., 2021; Lu et al., 2016).

Although so much evidence supporting the present findings, there are still some studies that contradict these results. Specifically, anthropomorphic design cues will enhance

users' expectations, which can result in frustration when these expectations are not realized (Nowak & Biocca, 2003). Further, Nowak (2004) indicated that low anthropomorphic images are more likable and credible than high anthropomorphic images and no anthropomorphic images are more attractive than anthropomorphic images. A possible explanation is that virtual agents with high anthropomorphic design cues lead to false expectations and disappointments (Knijnenburg & Willemsen, 2016; Schroeder, 2001). Users become frustrated when there is an imbalance between the anthropomorphism and actual capabilities of the virtual agent (Mimoun et al., 2012). Besides, another possible reason is that high anthropomorphic images will cause the uncanny valley effect, which may yield a sense of unease or even revulsion (Ciechanowski et al., 2019). However, previous research primarily focused on the anthropomorphic appearances of virtual agents. That is, users were only exposed to different appearances. In the present study, we took the whole chatbot interface as the research object. The proportion of the area of chatbot appearances only accounts for a small part of the interface. Apart from chatbot appearances, conversational styles were also considered in this research. This distinctive discrepancy may lead the present findings to be different from existing studies.

4.3. Combined measurement of the anthropomorphic effect

Noticeably, we observed significant effects of anthropomorphic design cues of chatbots on users' subjective perception and visual behaviors. Specifically, anthropomorphic appearances and human-like conversational styles jointly increased users' perception of the chatbot's social presence, trust in the chatbot, and satisfaction with the chatbot. Likewise, chatbots with anthropomorphic appearances could attract more fixation counts, longer average fixation duration, and longer dwell time. Besides, chatbots with human-like conversational styles attracted more fixation counts, shorter average fixation duration, and longer dwell time. These eye-tracking results suggested that chatbots with anthropomorphic design cues could attract more visual attention.

In fact, subjective evaluation results and eye-tracking outcomes are compatible and mutually supportive in this study. Our subjective evaluation results manifested that users' perceived social presence, perceived trust, and satisfaction are higher when a chatbot was designed with anthropomorphic visual appearance and human-like conversational style, which is also supported by social agency theory (Wiseman et al., 2012) that anthropomorphic cues could trigger users positive social responses. These positive social responses stimulate users' interest and result in more visual attention to the chatbot, which is ultimately reflected in eye-tracking outcomes. In turn, eye-tracking results manifested that anthropomorphic cues could attract more visual attention, which provided favorable support for subjective evaluation results in the present study. These results were also consistent with the media equation theory (Reeves & Nass, 1996)

that people instinctively view computer system as social actors during the interaction. Therefore, anthropomorphic design cues increased users' subjective perception and visual attention to chatbots.

5. Implications, limitations, and directions for future research

5.1. Implications

This study makes several significant theoretical contributions. Firstly, considering existing research primarily employed the self-reported method to investigate the influence of anthropomorphic appearances and conversational styles on users' subjective perception (Go & Sundar, 2019; Liebrecht et al., 2020), it would contribute to understanding the nature of the effect of these anthropomorphic cues on the chatbot design and revealing users' perception and visual behaviors. Next, compared with existing literature that has studied these two factors separately, the present study identified the joint effect of anthropomorphic appearances and human-like conversational styles on users' subjective perception. The results manifested that anthropomorphic appearances and conversational styles could induce users' higher perception of social presence, users' trust, and user satisfaction. Finally, to the best of the authors' knowledge, this is the first time to provide the combined method of subjective evaluation and eye-tracking technique to explore users' subjective perception and visual behaviors on the chatbot. It also provided a new research perspective for the chatbot design.

This study also provides valuable enlightenment to practitioners for the design of chatbots on the website. The present result showed that anthropomorphic appearances and conversational styles could enhance users' perception and visual attention. The findings signified that chatbot designers and researchers should consider incorporating anthropomorphic appearances and conversational styles in the chatbot. However, practitioners should be aware that anthropomorphism may lead to unexpected consequences. That is, high anthropomorphic design cues may create high expectations for users, which can result in disappointment or frustration when users' expectations are not balanced with the chatbots' capabilities (Knijnenburg & Willemsen, 2016).

5.2. Limitations, and directions for future research

This study also has limitations that can seed future inquiry. First, this research only explored the effect of anthropomorphic appearances and conversational styles on users' perception and visual behaviors. Beyond these two features, more anthropomorphic design cues, such as identity (Go & Sundar, 2019), humor (Niculescu et al., 2013), and communication delay (Schanke et al., 2021) may affect users' perception and visual behaviors. Therefore, future research could consider more anthropomorphic design cues in the chatbot design. Secondly, experiment stimuli were designed

to incorporate the experimental manipulation, which may not represent a typical chatbot on the website. Participants were told to imagine that they were communicating with a chatbot online. These measures increased internal validity, but future research can maximize ecological validity. Thirdly, within-subject experimental design was adopted in this study. Although we used a random presentation order of experimental stimuli, interference between different experimental conditions may affect the results. Still, this experiment design including randomization guarantees that the effect found were caused by our (within) manipulation and are not an order effect. Besides, participants' experimental tasks were also increased. Therefore, future research can consider the between-subject experimental design, which is considered as the "gold standard" in the domain of human-computer interaction research (Gergle & Tan, 2014).

6. Conclusion

The current study revealed whether and how anthropomorphic appearances and conversational styles affect users' subjective perception and visual behaviors on chatbots through the combined method of subjective evaluation and eye-tracking technique. The subjective evaluation results showed that anthropomorphic appearances and human-like conversational styles jointly enhanced users' perception of social presence, users' trust, and user satisfaction. Eye-tracking metrics indicated that chatbots with anthropomorphic appearances attracted more fixation counts, longer average fixation duration, and longer dwell time. Chatbots with human-like conversational styles attracted more fixation counts, shorter average fixation duration, and longer dwell time. These results suggested that anthropomorphic design cues could attract more visual attention to the chatbot. Additionally, these findings provided theoretical contributions for understanding the nature of the effects of anthropomorphic appearances and human-like conversational styles and valuable enlightenment to practitioners for the design of chatbots.

Acknowledgments

We would like to thank all the participants for their involvement in our research. Moreover, we are grateful to extend our gratitude to the editors and the reviewers for their valuable comments.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the National Natural Science Foundation of China [Grant No. 72071035].

ORCID

Jiahao Chen  <http://orcid.org/0000-0002-6110-7851>
Fu Guo  <http://orcid.org/0000-0001-6209-1874>

References

- Ahn, R. J., Cho, S. Y., & Sunny Tsai, W. (2022). Demystifying Computer-Generated Imagery (CGI) influencers: The effect of perceived anthropomorphism and social presence on brand outcomes. *Journal of Interactive Advertising*, 22(3), 327–335. <https://doi.org/10.1080/15252019.2022.2111242>
- Amin, A., Shah, B., Khattak, A. M., Moreira, F. J. L., Ali, G., Rocha, A., & Anwar, S. (2019). Cross-company customer churn prediction in telecommunication: A comparison of data transformation methods. *International Journal of Information Management*, 46, 304–319. <https://doi.org/10.1016/j.ijinfomgt.2018.08.015>
- Appel, J., von der Pütten, A., Krämer, N. C., & Gratch, J. (2012). Does humanity matter? Analyzing the importance of social cues and perceived agency of a computer system for the emergence of social reactions during human-computer interaction. *Advances in Human-Computer Interaction*, 2012(13), 1–10. <https://doi.org/10.1155/2012/324694>
- Araujo, T. (2018). Living up to the chatbot hype: The influence of anthropomorphic design cues and communicative agency framing on conversational agent and company perceptions. *Computers in Human Behavior*, 85, 183–189. <https://doi.org/10.1016/j.chb.2018.03.051>
- Ashfaq, M., Yun, J., Yu, S., & Loureiro, S. M. C. (2020). I, Chatbot: Modeling the determinants of users' satisfaction and continuance intention of AI-powered service agents. *Telematics and Informatics*, 54, 101473. <https://doi.org/10.1016/j.tele.2020.101473>
- Bartneck, C., Kulić, D., Croft, E., & Zoghbi, S. (2009). Measurement instruments for the anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety of Robots. *International Journal of Social Robotics*, 1(1), 71–81. <https://doi.org/10.1007/s12369-008-0001-3>
- Baylor, A. L., Rosenberg-Kima, R. B. (2006). Interface agents to alleviate online frustration. *Proceedings of the 7th international conference on learning sciences*. International Society of the Learning Sciences.
- Behe, B. K., Bae, M., Huddleston, P. T., & Sage, L. (2015). The effect of involvement on visual attention and product choice. *Journal of Retailing and Consumer Services*, 24, 10–21. <https://doi.org/10.1016/j.jretconser.2015.01.002>
- Benbasat, I., & Wang, W. (2005). Trust in and adoption of online recommendation agents. *Journal of the Association for Information Systems*, 6(3), 72–101. <https://doi.org/10.17705/1jais.00065>
- Bialkova, S., Grunert, K. G., & van Trijp, H. (2020). From desktop to supermarket shelf: Eye-tracking exploration on consumer attention and choice. *Food Quality and Preference*, 81, 103839. <https://doi.org/10.1016/j.foodqual.2019.103839>
- Boardman, R., McCormick, H., & Henninger, C. E. (2022). Exploring attention on a retailer's homepage: An eye-tracking & qualitative research study. *Behaviour & Information Technology*, 1–17. <https://doi.org/10.1080/0144929X.2022.2059396>
- Borau, S., Otterbring, T., Laporte, S., & Fosso Wamba, S. (2021). The most human bot: Female gendering increases humanness perceptions of bots and acceptance of AI. *Psychology & Marketing*, 38(7), 1052–1068. <https://doi.org/10.1002/mar.21480>
- Burgoon, J. K., & Walther, J. B. (1990). Nonverbal expectancies and the evaluative consequences of violations. *Human Communication Research*, 17(2), 232–265. <https://doi.org/10.1111/j.1468-2958.1990.tb00232.x>
- Cao, Y., Ding, Y., Proctor, R. W., Duffy, V. G., Liu, Y., & Zhang, X. (2021). Detecting users' usage intentions for websites employing deep learning on eye-tracking data. *Information Technology and Management*, 22(4), 281–292. <https://doi.org/10.1007/s10799-021-00336-6>
- Cao, Y., Zhang, Y., Ding, Y., Duffy, V. G., & Zhang, X. (2021). Is an anthropomorphic app icon more attractive? Evidence from neuroergonomics. *Applied Ergonomics*, 97, 103545. <https://doi.org/10.1016/j.apergo.2021.103545>
- Chattaraman, V., Kwon, W.-S., Gilbert, J. E., & Li, Y. (2014). Virtual shopping agents: Persona effects for older users. *Journal of Research in Interactive Marketing*, 8(2), 144–162. <https://doi.org/10.1108/JRIM-08-2013-0054>

- Chaves, A. P., & Gerosa, M. A. (2021). How should my chatbot interact? A survey on social characteristics in human-Chatbot interaction design. *International Journal of Human-Computer Interaction*, 37(8), 729–758. <https://doi.org/10.1080/10447318.2020.1841438>
- Chung, M., Ko, E., Joung, H., & Kim, S. J. (2020). Chatbot e-service and customer satisfaction regarding luxury brands. *Journal of Business Research*, 117, 587–595. <https://doi.org/10.1016/j.jbusres.2018.10.004>
- Ciechanowski, L., Przegalinska, A., Magnuski, M., & Gloor, P. (2019). In the shades of the uncanny valley: An experimental study of human-Chatbot interaction. *Future Generation Computer Systems*, 92, 539–548. <https://doi.org/10.1016/j.future.2018.01.055>
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences*. Academic Press. <https://doi.org/10.1016/B978-0-12-179060-8.50012-8>
- Cyr, D., Hassanein, K., Head, M., & Ivanov, A. (2007). The role of social presence in establishing loyalty in e-service environments. *Interacting with Computers*, 19(1), 43–56. <https://doi.org/10.1016/j.intcom.2006.07.010>
- Egner, T. (2007). Congruency sequence effects and cognitive control. *Cognitive, Affective, & Behavioral Neuroscience*, 7(4), 380–390. <https://doi.org/10.3758/CABN.7.4.380>
- Etamad-Sajadi, R. (2016). The impact of online real-time interactivity on patronage intention: The use of avatars. *Computers in Human Behavior*, 61, 227–232. <https://doi.org/10.1016/j.chb.2016.03.045>
- Feine, J., Gnewuch, U., Morana, S., & Maedche, A. (2019). A taxonomy of social cues for conversational agents. *International Journal of Human-Computer Studies*, 132, 138–161. <https://doi.org/10.1016/j.ijhcs.2019.07.009>
- Følstad, A., & Brandtzaeg, P. B. (2020). Users' experiences with Chatbots: Findings from a questionnaire study. *Quality and User Experience*, 5(1), 1–14. <https://doi.org/10.1007/s41233-020-00033-2>
- Følstad, A., Nordheim, C. B., Bjørkli, C. A. (2018). What makes users trust a chatbot for customer service? An exploratory interview study. *International conference on internet science*, 194–208. https://doi.org/10.1007/978-3-030-01437-7_16
- Gefen, D., & Straub, D. (2003). Managing user trust in B2C e-services. *e-Service*, 2(2), 7–24. <https://doi.org/10.2979/esj.2003.2.2.7>
- Gergle, D., & Tan, D. S. (2014). Experimental research in HCI. In *Ways of Knowing in HCI*, 191–227. https://doi.org/10.1007/978-1-4939-0378-8_9
- Glaholt, M. G., Wu, M.-C., & Reingold, E. M. (2009). Predicting preference from fixations. *PsychNology J*, 7(2), 141–158.
- Gnewuch, U., Morana, S., Adam, M., & Maedche, A. (2018). Faster is not always better: Understanding the effect of dynamic response delays in human-chatbot interaction. *European Conference on Information Systems (ECIS2018)*.
- Go, E., & Sundar, S. S. (2019). Humanizing chatbots: The effects of visual, identity and conversational cues on humanness perceptions. *Computers in Human Behavior*, 97, 304–316. <https://doi.org/10.1016/j.chb.2019.01.020>
- Gong, L., & Nass, C. (2007). When a talking-face computer agent is half-human and half-humanoid: Human identity and consistency preference. *Human Communication Research*, 33(2), 163–193. <https://doi.org/10.1111/j.1468-2958.2007.00295.x>
- Guo, F., Chen, J., Li, M., Lyu, W., & Zhang, J. (2022). Effects of visual complexity on user search behavior and satisfaction: An eye-tracking study of mobile news apps. *Universal Access in the Information Society*, 21(4), 795–808. <https://doi.org/10.1007/s10209-021-00815-1>
- Guo, F., Ding, Y., Liu, W., Liu, C., & Zhang, X. (2016). Can eye-tracking data be measured to assess product design?: Visual attention mechanism should be considered. *International Journal of Industrial Ergonomics*, 53, 229–235. <https://doi.org/10.1016/j.ergon.2015.12.001>
- Guo, K., Tunnicliffe, D., & Roebuck, H. (2010). Human spontaneous gaze patterns in viewing of faces of different species. *Perception*, 39(4), 533–542. <https://doi.org/10.1068/p6517>
- Guo, F., & Zhang, X. (2020). The impact of brand history on consumers' cognitive process and brand attitude. *Journal of Neuroscience, Psychology, and Economics*, 13(4), 191–203. <https://doi.org/10.1037/npe0000136>
- Hassanein, K., & Head, M. (2007). Manipulating perceived social presence through the web interface and its impact on attitude towards online shopping. *International Journal of Human-Computer Studies*, 65(8), 689–708. <https://doi.org/10.1016/j.ijhcs.2006.11.018>
- Ho, C.-H., & Hou, K.-C. (2015). Exploring the attractive factors of app icons. *KSI Transactions on Internet and Information Systems (TIIS)*, 9(6), 2251–2270. <https://doi.org/10.3837/tiis.2015.06.016>
- Holmqvist, K., Nyström, M., Andersson, R., Dewhurst, R., Jarodzka, H., & Van de Weijer, J. (2011). *Eye tracking: A comprehensive guide to methods and measures*. OUP Oxford.
- Jackson, R. B., & Williams, T. (2021). A theory of social agency for human-robot interaction. *Frontiers in Robotics and AI*, 8, 687726. <https://doi.org/10.3389/frobt.2021.687726>
- Jacob, R. J., & Karn, K. S. (2003). Eye tracking in human-computer interaction and usability research: Ready to deliver the promises. In *The Mind's Eye* (pp. 573–605). Elsevier. <https://doi.org/10.1016/B978-044451020-4/50031-1>
- Jarodzka, H., Skuballa, I., & Gruber, H. (2021). Eye-tracking in educational practice: Investigating visual perception underlying teaching and learning in the classroom. *Educational Psychology Review*, 33(1), 1–10. <https://doi.org/10.1007/s10648-020-09565-7>
- Jiang, H., Cheng, Y., Yang, J., & Gao, S. (2022). AI-powered chatbot communication with customers: Dialogic interactions, satisfaction, engagement, and customer behavior. *Computers in Human Behavior*, 134, 107329. <https://doi.org/10.1016/j.chb.2022.107329>
- Just, M. A., & Carpenter, P. A. (1976). Eye fixations and cognitive processes. *Cognitive Psychology*, 8(4), 441–480. [https://doi.org/10.1016/0010-0285\(76\)90015-3](https://doi.org/10.1016/0010-0285(76)90015-3)
- Klein, K., & Martinez, L. F. (2022). The impact of anthropomorphism on customer satisfaction in chatbot commerce: An experimental study in the food sector. *Electronic Commerce Research*, 1–37. <https://doi.org/10.1007/s10660-022-09562-8>
- Knijnenburg, B. P., & Willemsen, M. C. (2016). Inferring capabilities of intelligent agents from their external traits. *ACM Transactions on Interactive Intelligent Systems*, 6(4), 1–25. <https://doi.org/10.1145/2963106>
- Lee, K. M. (2004). Presence, explicated. *Communication Theory*, 14(1), 27–50. <https://doi.org/10.1111/j.1468-2885.2004.tb00302.x>
- Lee, K. M. (2008). Media equation theory. In *The International Encyclopedia of Communication*. <https://doi.org/10.1002/9781405186407.wbiecm035>
- Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1), 50–80. https://doi.org/10.1518/hfes.46.1.50_30392
- Letheren, K., & Glavas, C. (2017). Embracing the bots: How direct to consumer advertising is about to change forever. *The Conversation*.
- Li, M., Guo, F., Fang, C., & Li, F. (2022). Multisensory integration effect of humanoid robot appearance and voice on users' affective preference and visual attention. *Behaviour & Information Technology*, 1–20. <https://doi.org/10.1080/0144929X.2022.2125830>
- Li, M., Guo, F., Ren, Z., & Duffy, V. G. (2022). A visual and neural evaluation of the affective impression on humanoid robot appearances in free viewing. *International Journal of Industrial Ergonomics*, 88, 103159. <https://doi.org/10.1016/j.ergon.2021.103159>
- Li, M., & Wang, R. (2023). Chatbots in e-commerce: The effect of chatbot language style on customers' continuance usage intention and attitude toward brand. *Journal of Retailing and Consumer Services*, 71. <https://doi.org/10.1080/0144929X.2022.2125830>
- Liebrecht, C., Sander, L., & Hooijdonk, C. v (2020). Too informal? How a chatbot's communication style affects brand attitude and quality of interaction. *International Workshop on Chatbot Research and Design*, 16–31. https://doi.org/10.1007/978-3-030-68288-0_2
- Liebrecht, C., Tsaousi, C., & van Hooijdonk, C. (2021). Linguistic elements of conversational human voice in online brand communication: Manipulations and perceptions. *Journal of Business Research*, 132, 124–135. <https://doi.org/10.1016/j.jbusres.2021.03.050>
- Liebrecht, C., & van Hooijdonk, C. (2020). Creating humanlike chatbots: What chatbot developers could learn from webcare employees in adopting a conversational human voice. In *International*

- Workshop on Chatbot Research and Design, (pp. 51–64). https://doi.org/10.1007/978-3-030-39540-7_4
- Lu, B., Fan, W., & Zhou, M. (2016). Social presence, trust, and social commerce purchase intention: An empirical research. *Computers in Human Behavior*, 56, 225–237. <https://doi.org/10.1016/j.chb.2015.11.057>
- Luan, J., Yao, Z., Zhao, F., & Liu, H. (2016). Search product and experience product online reviews: An eye-tracking study on consumers' review search behavior. *Computers in Human Behavior*, 65, 420–430. <https://doi.org/10.1016/j.chb.2016.08.037>
- Lui, A., & Lamb, G. W. (2018). Artificial intelligence and augmented intelligence collaboration: Regaining trust and confidence in the financial sector. *Information & Communications Technology Law*, 27(3), 267–283. <https://doi.org/10.1080/13600834.2018.1488659>
- Martin, N., & Morich, K. (2011). Unconscious mental processes in consumer choice: Toward a new model of consumer behavior. *Journal of Brand Management*, 18(7), 483–505. <https://doi.org/10.1057/bm.2011.10>
- Meyer, J., Miller, C., Hancock, P., De Visser, E. J., Dorneich, M. (2016). Politeness in machine-human and human-human interaction. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 60(1), 279–283. <https://doi.org/10.1177/1541931213601064>
- Mimoun, M. S. B., Poncin, I., & Garnier, M. (2012). Case study – Embodied virtual agents: An analysis on reasons for failure. *Journal of Retailing and Consumer Services*, 19(6), 605–612. <https://doi.org/10.1016/j.jretconser.2012.07.006>
- Mostafa, R. B., & Kasamani, T. (2022). Antecedents and consequences of chatbot initial trust. *European Journal of Marketing*, 56(6), 1748–1771. <https://doi.org/10.1108/ejm-02-2020-0084>
- Muralidharan, L., de Visser, E. J., & Parasuraman, R. (2014). The effects of pitch contour and flanging on trust in speaking cognitive agents. In *CHI'14 Extended Abstracts on Human Factors in Computing Systems*, 2167–2172. <https://doi.org/10.1145/2559206.2581231>
- Nass, C., Fogg, B., & Moon, Y. (1996). Can computers be teammates? *International Journal of Human-Computer Studies*, 45(6), 669–678. <https://doi.org/10.1006/ijhc.1996.0073>
- Nass, C., & Moon, Y. (2000). Machines and mindlessness: Social responses to computers. *Journal of Social Issues*, 56(1), 81–103. <https://doi.org/10.1111/0022-4537.00153>
- Niculescu, A., van Dijk, B., Nijholt, A., Li, H., & See, S. L. (2013). Making social robots more attractive: The effects of voice pitch, humor and empathy. *International Journal of Social Robotics*, 5(2), 171–191. <https://doi.org/10.1007/s12369-012-0171-x>
- Nowak, K. L. (2004). The influence of anthropomorphism and agency on social judgment in virtual environments. *Journal of Computer-Mediated Communication*, 9(2), JCMC925. <https://doi.org/10.1111/j.1083-6101.2004.tb00284.x>
- Nowak, K. L., & Biocca, F. (2003). The effect of the agency and anthropomorphism on users' sense of telepresence, copresence, and social presence in virtual environments. *Presence*, 12(5), 481–494. <https://doi.org/10.1162/105474603322761289>
- Nowak, K. L., & Rauh, C. (2005). The influence of the avatar on online perceptions of anthropomorphism, androgyny, credibility, homophily, and attraction. *Journal of Computer-Mediated Communication*, 11(1), 153–178. <https://doi.org/10.1111/j.1083-6101.2006.tb00308.x>
- Oh, C. S., Bailenson, J. N., & Welch, G. F. (2018). A systematic review of social presence: Definition, antecedents, and implications. *Frontiers in Robotics and AI*, 5, 114. <https://doi.org/10.3389/frobt.2018.00114>
- Oliver, R. L. (1980). A cognitive model of the antecedents and consequences of satisfaction decisions. *Journal of Marketing Research*, 17(4), 460–469. <https://doi.org/10.1177/002224378001700405>
- Paulhus, L. D., & Vazire, S. (2007). The Self-report Method. In R.W. Robins, R. C. Fraley, & R. F. Krueger (Eds.), *Handbook of Research Methods in Personality Psychology*, (pp. 224–239). New York, NY: The Guilford Press.
- Pawlik, V. P. (2022). Design matters! how visual gendered anthropomorphic design cues moderate the determinants of the behavioral intention towards using chatbots. In *Chatbot Research and Design*, 192–208. https://doi.org/10.1007/978-3-030-94890-0_12
- Pei, H., Huang, X., & Ding, M. (2022). Image visualization: Dynamic and static images generate users' visual cognitive experience using eye-tracking technology. *Displays*, 73, 102175. <https://doi.org/10.1016/j.displa.2022.102175>
- Qiu, L., & Benbasat, I. (2009). Evaluating anthropomorphic product recommendation agents: A social relationship perspective to designing information systems. *Journal of Management Information Systems*, 25(4), 145–182. <https://doi.org/10.2753/MIS0742-1222250405>
- Qu, Q.-X., & Guo, F. (2019). Can eye movements be effectively measured to assess product design? Gender differences should be considered. *International Journal of Industrial Ergonomics*, 72, 281–289. <https://doi.org/10.1016/j.ergon.2019.06.006>
- Ramos Gameiro, R., Kaspar, K., König, S. U., Nordholt, S., & König, P. (2017). Exploration and exploitation in natural viewing behavior. *Scientific Reports*, 7(1), 1–23. <https://doi.org/10.1038/s41598-017-02526-1>
- Reeves, B., & Nass, C. (1996). *The media equation: How people treat computers, television, and new media like real people* (vol. 10, p. 236605) Cambridge University Press.
- Rhim, J., Kwak, M., Gong, Y., & Gweon, G. (2022). Application of humanization to survey chatbots: Change in chatbot perception, interaction experience, and survey data quality. *Computers in Human Behavior*, 126, 107034. <https://doi.org/10.1016/j.chb.2021.107034>
- Roy, R., & Naidoo, V. (2021). Enhancing chatbot effectiveness: The role of anthropomorphic conversational styles and time orientation. *Journal of Business Research*, 126, 23–34. <https://doi.org/10.1016/j.jbusres.2020.12.051>
- Schanke, S., Burtch, G., & Ray, G. (2021). Estimating the impact of “humanizing” customer service Chatbots. *Information Systems Research*, 32(3), 736–751. <https://doi.org/10.1287/isre.2021.1015>
- Schroeder, R. (2001). *The social life of avatars: Presence and interaction in shared virtual environments*. Springer Science & Business Media.
- Schuetzler, R. M., Grimes, G. M., & Scott Giboney, J. (2020). The impact of chatbot conversational skill on engagement and perceived humanness. *Journal of Management Information Systems*, 37(3), 875–900. <https://doi.org/10.1080/07421222.2020.1790204>
- Seitz, L., Bekmeier-Feuerhahn, S., & Gohil, K. (2022). Can we trust a chatbot like a physician? A qualitative study on understanding the emergence of trust toward diagnostic chatbots. *International Journal of Human-Computer Studies*, 165, 102848. <https://doi.org/10.1016/j.ijhcs.2022.102848>
- Song, S. W., & Shin, M. (2022). Uncanny valley effects on Chatbot Trust, purchase intention, and adoption intention in the context of E-Commerce: The moderating role of avatar familiarity. *International Journal of Human-Computer Interaction*, 1–16. <https://doi.org/10.1080/10447318.2022.2121038>
- Spelt, H. A. A., Westerink, J. H. D. M., Ham, J., & IJsselstein, W. A. (2019). Psychophysiological reactions to persuasive messages deploying persuasion principles. *IEEE Transactions on Affective Computing*, 13(1), 461–472. <https://doi.org/10.1109/TAFFC.2019.2931689>
- Tinwell, A. (2009). Uncanny as usability obstacle. *International Conference on Online Communities and Social Computing*, 622–631. https://doi.org/10.1007/978-3-642-02774-1_67
- Tsai, W.-H. S., Liu, Y., & Chuan, C.-H. (2021). How chatbots' social presence communication enhances consumer engagement: The mediating role of parasocial interaction and dialogue. *Journal of Research in Interactive Marketing*, 15(3), 460–482. <https://doi.org/10.1108/JRIM-12-2019-0200>
- Unema, P. J., Pannasch, S., Joos, M., & Velichkovsky, B. M. (2005). Time course of information processing during scene perception: The relationship between saccade amplitude and fixation duration. *Visual Cognition*, 12(3), 473–494. <https://doi.org/10.1080/1350628044000409>
- Verhagen, T., Van Nes, J., Feldberg, F., & Van Dolen, W. (2014). Virtual customer service agents: Using social presence and personalization to shape online service encounters. *Journal of Computer-Mediated Communication*, 19(3), 529–545. <https://doi.org/10.1111/jcc4.12066>

- Wang, L. C., Baker, J., Wagner, J. A., & Wakefield, K. (2007). Can a retail web site be social? *Journal of Marketing*, 71(3), 143–157. <https://doi.org/10.1509/jmkg.71.3.143>
- Wang, X-s., Guo, F., Li, M-m., & Tian, X-h (2021). Effects of mobile news interface design features on users' gaze behaviours and behavioural performance: Evidence from China. *Behaviour & Information Technology*, 40(16), 1741–1754. <https://doi.org/10.1080/0144929x.2020.1778089>
- Wang, Q., Yang, Y., Wang, Q., & Ma, Q. (2014). The effect of human image in B2C website design: An eye-tracking study. *Enterprise Information Systems*, 8(5), 582–605. <https://doi.org/10.1080/17517575.2014.925585>
- Wiseman, R. M., Cuevas-Rodríguez, G., & Gomez-Mejia, L. R. (2012). Towards a social theory of agency. *Journal of Management Studies*, 49(1), 202–222. <https://doi.org/10.1111/j.1467-6486.2011.01016.x>
- Xie, C., Wang, Y., & Cheng, Y. (2022). Does artificial intelligence satisfy you? A meta-analysis of user gratification and user satisfaction with AI-powered Chatbots. *International Journal of Human-Computer Interaction*, 1–11. <https://doi.org/10.1080/10447318.2022.2121458>
- Xu, Y., Zhang, J., Chi, R., & Deng, G. (2022). Enhancing customer satisfaction with chatbots: The influence of anthropomorphic communication styles and anthropomorphised roles. *Nankai Business Review International*, 1–23. <https://doi.org/10.1108/NBRI-06-2021-0041>
- Zhou, X., Kim, S., & Wang, L. (2019). Money helps when money feels: Money anthropomorphism increases charitable giving. *Journal of Consumer Research*, 45(5), 953–972. <https://doi.org/10.1093/jcr/ucy012>

About the authors

Jiahao Chen is a PhD student at the Department of Industrial Engineering, School of Business Administration, Northeastern

University, China. His research interests include human factors, human-chatbot interaction, user experience design, and human-computer interaction.

Fu Guo is a professor of Industrial Engineering at School of Business Administration, Northeastern University, China. Her research interests include human factors, Kansei engineering, user experience, human-computer interaction, human-robot interaction, occupational safety and health, and product placement.

Zenggen Ren is a PhD student at the Department of Industrial Engineering, School of Business Administration, Northeastern University, China. His research interests include human factors, Kansei engineering, and human-robot interaction.

Mingming Li is a PhD candidate at the Department of Industrial Engineering, School of Business Administration, Northeastern University, China. His research interests include human factors, human-robot interaction, user experience design, and human-computer interaction.

Jaap Ham is an associated professor in the research group of Human Technology Interaction, Department of Industrial Engineering and Innovation Sciences, Eindhoven University of Technology, The Netherlands. He focuses on ambient forms of technology and social forms of technology (such as social robots) to influence people, health behavior, and sustainability.

Appendix A



High anthropomorphic appearances and human-like styles

High anthropomorphic appearances and mechanical styles



Medium anthropomorphic appearances and human-like styles

Medium anthropomorphic appearances and mechanical styles



Low anthropomorphic appearances and human-like styles

Low anthropomorphic appearances and mechanical styles

Examples of experimental stimuli.